REACTION TIME

EMG

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10M+

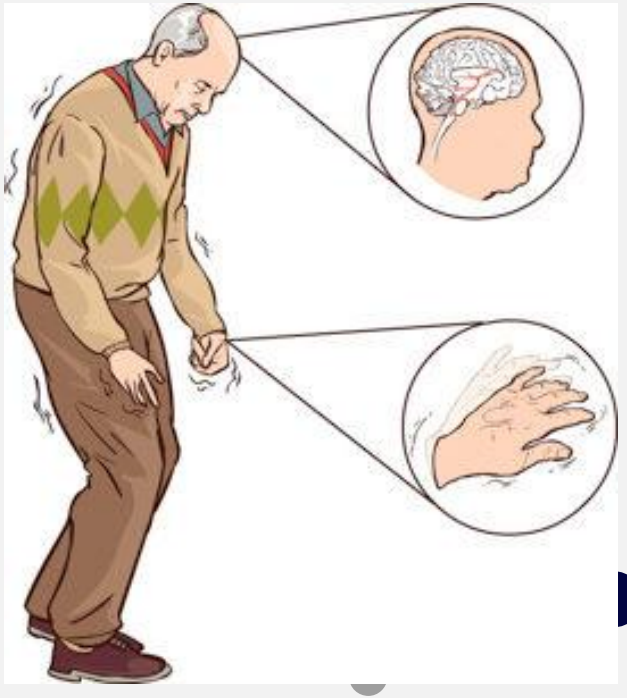
People worldwide are living with Parkinson's disease. By **2030**, an estimated **1.2 million people** in the **U.S.** alone will have it.

Why Measure Reaction Time?

Parkinson's patients show **delayed motor initiation** and **abnormal muscle activation**, leading to **longer**, more **variable reaction times**.

Compared Reaction Times

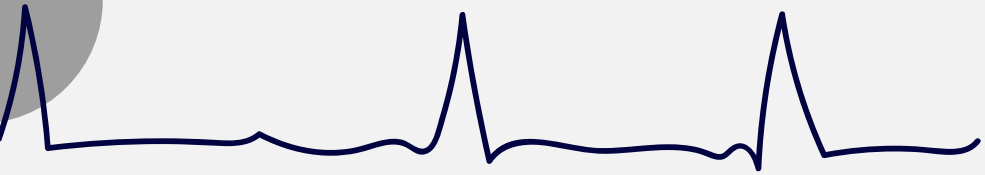
- Parkinson’s Patient’s avg reaction time: 300 - 500 ms
- Healthy person’s avg reaction time: 200 - 300 ms



Physiologic Requirements to Engineering Goals

SNR

Magnitude of signal: muscle contraction (1 - 300 μV)
Noise Sources: thermal, patient nervousness, environmental vibrations, EMI, heartbeat



Bandwidth

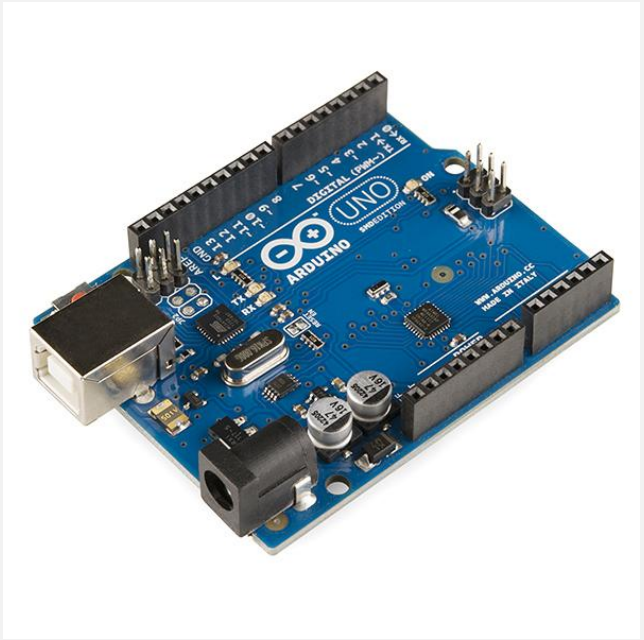
Range of muscle contraction, indicative of neurological disorder

Physiologic Requirements to Engineering Goals Cont.

Power and Cost

±9V battery powered for patient isolation. Arduino Uno 10-bit ADC is portable and requires no calibration.

Total cost: ~$50

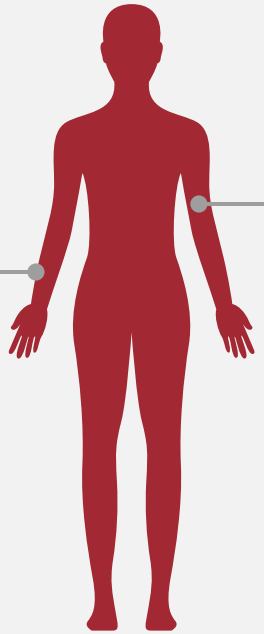


Measuring Reaction Time w/ EMGs

Needle EMG

- Inserted into muscles
- Standard for diagnosing neuromuscular disorders
- **Invasive, painful,** and **requires** a **trained clinician**





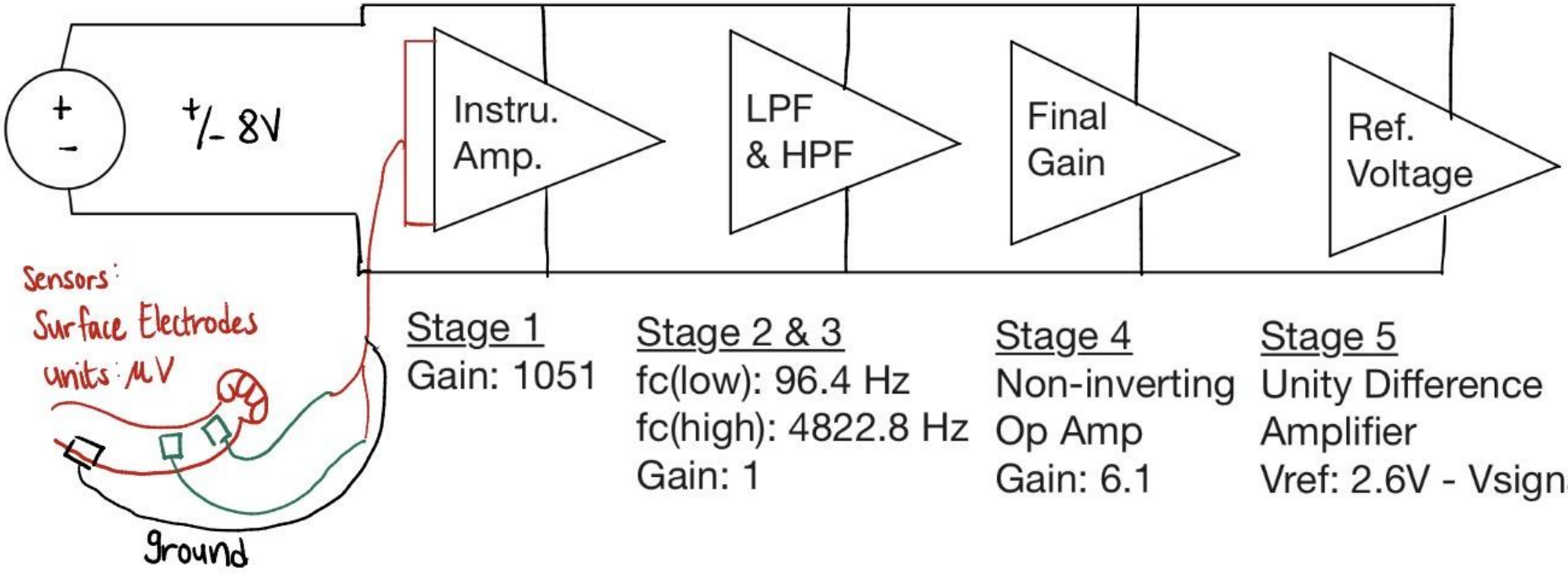
Surface EMG

- Ag/AgCl electrodes placed on the skin
- Ideal for timing muscle activation onset
- **Weaker signal and resolution**



EMG Circuit

Estimated physiological
BW: 96.4 Hz - 4.82 kHz



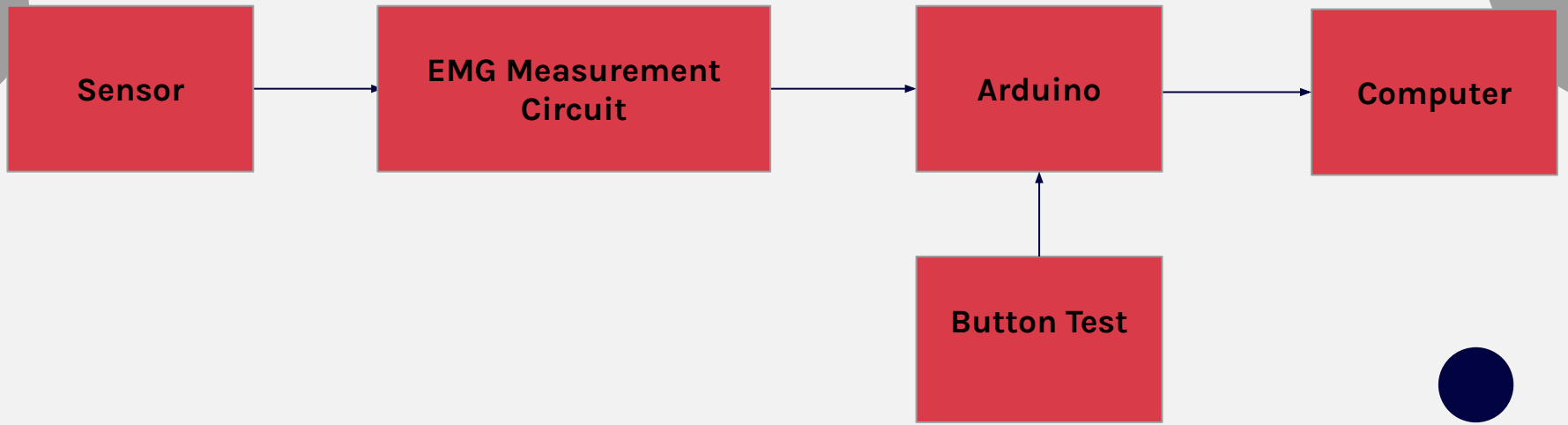
Stage 1
Gain: 1051

Stage 2 & 3
fc(low): 96.4 Hz
fc(high): 4822.8 Hz
Gain: 1

Stage 4
Non-inverting Op Amp
Gain: 6.1

Stage 5
Unity Difference Amplifier
Vref: 2.6V - Vsignal

Data Flow Circuit



Signal, Noise & Interference Model

SNR

- Signal (flex): 1.65 V above 0 V baseline
- Noise RMS (rest): ~0.04 V
- SNR ≈ 32 dB

60 Hz EMI

- HPF Cutoff = 96.4 Hz → 60 Hz in stopband
- INA CMRR ~80–100 dB rejects power-line interference

Optimal Noise Matching

- Source impedance ~5 kΩ
- Voltage noise (8 nV/√Hz) dominates current noise → well matched
- No external noise-matching network needed

Anti-aliasing Filter & ADC

AAF Design

AAF Cutoff: ~4800 Hz (LPF, Stage 2/3)

Sampling Rate

Nyquist requires > ~9,600 Sps

Arduino Uno R3 ADC: 9,615 Sps ✓

ADC: 10-bits, 0-5 V range

- Resolution: LSB = 4.88 mV/step
- Quantization SNR: ~62 dB >> signal SNR (32 dB)
- Quantization noise not limiting factor

Measuring Reaction Time

Button Pressing Test

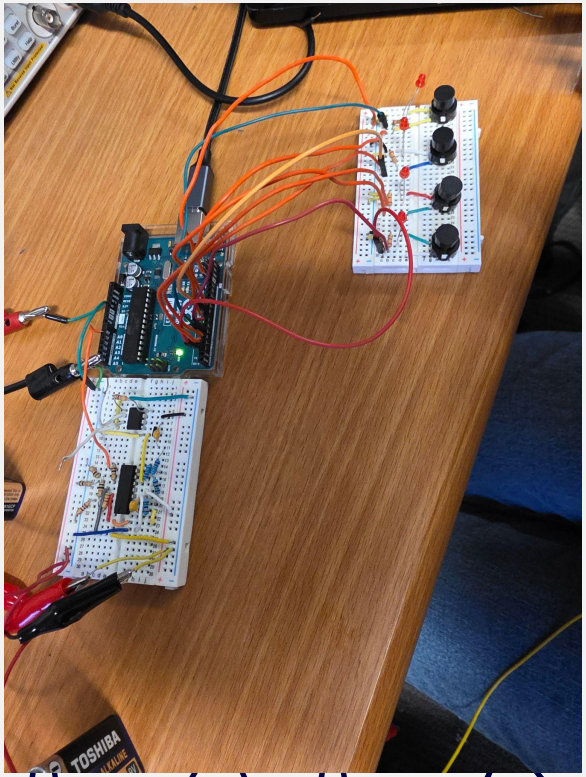
Mini-game to press the right button

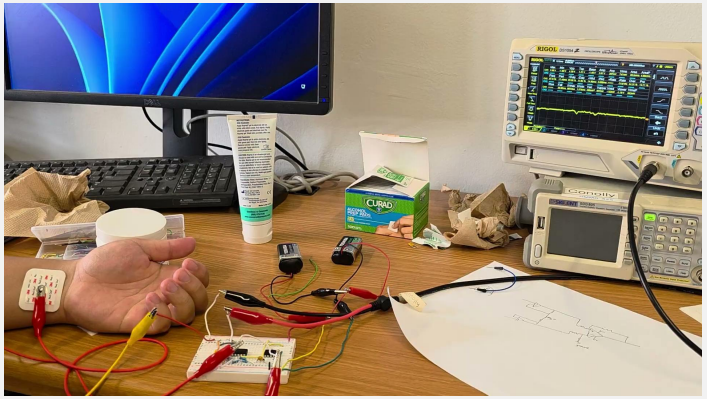
2 TIMERS:

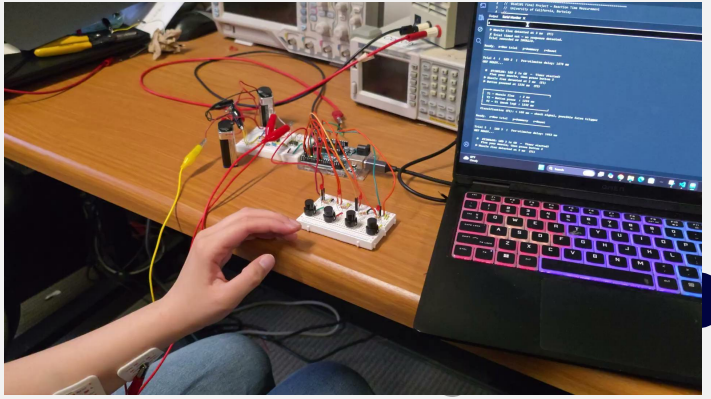
- Threshold of signal amplitude as EMG timer
- Button timer approx 100 to 200 ms slower than signal (assuming the person hits the right button)



Physical Design + Demo







Prototype vs. Gold Standard

Our Prototype

Trial	1	2	3	4	5	6
Muscle flex time T1 (ms)	266	316	221	218	224	269
Button press time T2 (ms)	836	540	432	965	806	580
Mechanical lag T2-T1 (ms)	570	224	211	747	582	311

Mean T1: 252 ± 35 ms

Mean T2: 693 ms ± 188 ms

Mean T2-T1: 441 ms ± 203 ms

Prototype vs. Gold Standard Cont.

Gold Standard

- Gold standard reaction time: 200 - 300 ± 10 ms
- Our result: 252 ± 35 ms (25 ms above gold standard)
- Pros: Within normal healthy range, non-invasive, ~$50 cost
- Cons: Higher variability, motion artifacts affect threshold

Tradeoffs & Lessons Learned

01

Instability:
Slight movement disorients components → easily detached and broken circuit → difficult to repair

02

Lost Signal to avoid EMI:
Lower cutoff frequency from 10 Hz → 100 Hz
Avoid 60 Hz EMI and heartbeat. We lost signal but lowered introduced noise.

03

Frequent Backtracking:
Sanity checks are **vital!** Constantly test, debug and backtrack!

Conclusions

- **Potential supplemental screening tool** for Parkinson's and other neurodegenerative disorders
- **Low-cost, non-invasive** way to measure reaction time with EMG
- Splitting gain and filtering in between **amplifies** a **biopotential signal without amplifying noise** and ensures the integrity of signal
- **Future reference** = solder all connections and test each phase as it is added

THANK YOU

For listening

References

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